

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

Roll No. :S040	Name:ADITYA SOPTE
Class :SY IT	Batch :
Date of Assignment :12-08-2026	Date/Time of Submission :12-08-2026

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

CODE:

```
vc.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/vc.py (3.14.6)
File Edit Format Run Options Window Help
def bubbleSort(array):
    for i in range(len(array)):
        for j in range(0, len(array) - i - 1):
            if array[j] > array[j + 1]:
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp
data = [-2, 45, 0, 11, -9]
bubbleSort(data)
print("Sorted Array in Ascending Order")
print(data)
print("Aditya Sopte S040")
```

OUTPUT:

```
Sorted Array in Ascending Order
[-9, -2, 0, 11, 45]
Aditya Sopte S040
>>>|
```

b. Insertion sort

CODE:

```
vc.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/vc.py (3.14.6)
File Edit Format Run Options Window Help
def insertionSort(array):
    for step in range(1, len(array)):
        key = array[step]
        j = step - 1
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1
        array[j + 1] = key
data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("Aditya Sopte S040")
```

OUTPUT:

```
===== RESTART: C:/
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
Aditya Sopte S040
>>>
```

c. Selection sort

CODE:

```
def selectionSort(array, size):  
  
    for step in range(size):  
        min_idx = step  
  
        for i in range(step + 1, size):  
            if array[i] < array[min_idx]:  
                min_idx = i  
  
        (array[step], array[min_idx]) = (array[min_idx], array[step])  
  
data = [-2, 45, 0, 11, -9]  
size = len(data)  
selectionSort(data, size)  
print("Sorted Array in Ascending Order:")  
print(data)  
print("Aditya Sopte S040")  
|
```

OUTPUT:

```
===== RESTART: C:  
Sorted Array in Ascending Order:  
[-9, -2, 0, 11, 45]  
Aditya Sopte S040  
>>>
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in descending order using **Bubble Sort**.
2. Write a Python program to sort a list of numbers in descending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in descending order using **Selection Sort**.

Theory Questions

4. What is sorting? Why is it required?
5. What is the basic idea of Bubble Sort?
6. How does Insertion Sort work?
7. How does Selection Sort work?
8. What is a **pass** in a sorting algorithm?
9. What is the difference between Bubble Sort and Selection Sort?
10. Which sorting algorithm is suitable when the data is already nearly sorted? Why?